

General Industries Hub

Enterprise Inventory Management

Store Keeper Manual

Site-level material handling, daily entries, returnables.

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Confidential - for authorized personnel only

GI Hub - Role booklet - Store Keeper Manual

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1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a multi-site warehouse inventory, procurement and asset-management system. It tracks every material movement - issues, receipts, returns and corrections - through a two-stage approval process, and gives you live stock figures, oldest-stock-first discipline on anything with an expiry date, stock valuation, serialised tool and equipment tracking, and a complete audit trail across every site you manage.

It runs in a web browser on a phone, a tablet or a desktop, and is also available as an installable app for Windows, macOS and Android.

1.2 Core principles

Principle	What it means
Stock is calculated, never stored	Stock on hand is always received, minus consumed, minus returned, recalculated from the movement history every time it is shown. No running total is kept anywhere, so no
Two-stage approval	Store Keepers stage entries step-HOD ready and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal
Audit-first	Every system action is recorded with the user's name, the exact time and what changed. Even a deletion leaves a trace.
Nothing to maintain	Upgrades apply themselves when the system starts. Nobody has to run a database update by hand.
Procurement chain	One continuous workflow from a site's request through to material on the shelf. Logistics owns purchase orders; Warehouse owns physical receiving and delivery notes; the Site HOD approves what is being sent; the Store Keeper
RL/BL strict separation	Rubber Lining and Brick Lining items NEVER share a PO line group, a DN, or a warehouse aggregation. The system rejects mixed-family DNs by design and tags each line with its family on insert.

Warehouse-blind pricing	Warehouse users can see materials and quantities but NEVER see Unit_Price, Total_Price, or any monetary header field on a PO. Three independent enforcement layers guarantee this.
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1.3 The transaction lifecycle (the heart of the system)

Every stock movement travels the same four steps. Nothing reaches the permanent record until a Head of Department has approved it.

Step	Who does it	What happens	Where it sits afterwards
1	Store Keeper	Enters the issue, receipt or return on the Entry Log	Staged, awaiting approval
2	Head of Department	Reviews it at End-of-Day and may edit the quantity, approve, or reject with a reason	Still staged until they decide
3	System	Commits the approved entry to the permanent stock	Permanent - stock figures move now
4	System	Sends Sends the WhatsApp and in-app notifications, and writes the audit entry	Permanent audit history

Receipts, issues, returns and stock adjustments all follow this same shape. A rejected entry never reaches step 3, so it never affects a stock figure - but the rejection itself is still recorded, together with the reason.

1.3a The procurement chain

Buying material is a relay. Each role does one thing and hands over to the next, and every handover appears in the receiving role's own queue.

Step	Role	Action	Lands in
1	Site HOD	Submits a Purchase Request	Logistics - Incoming PRs
2	Logistics	Issues the Purchase Order	Site HOD is notified; Admin sees it for oversight
3	Logistics	Assigns the order (or individual lines) to a warehouse	Warehouse - Incoming Assignments
4	Warehouse	Acknowledges, then records the physical arrival from	Warehouse - Receiving
5	Warehouse	Prepares the Delivery Note	Logistics - DN approval queue
6	Logistics	Approves the delivery date	Site HOD - DN Approvals

Step	Role	Action	Lands in
7	Site HOD	Approves the contents	Store Keeper - Incoming DNs
8	Store Keeper	Confirms physical arrival on	Permanent stock record; the

Three side-paths exist for when things do not go to plan: site delivery closes

- Vendor returns - any role can raise one. The order line reopens so the material can be re-delivered.
- Reschedules - Warehouse or Site HOD request a new date; Logistics decides.
- Force-closure - Logistics only, for an order that will never complete. Both Admin and the Site HOD are notified automatically.

A Delivery Note may never mix Rubber Lining and Brick Lining material. The system refuses to create one, rather than warning about it.

1.4 Currency, dates, units

- Currency: Saudi Riyal (SAR) throughout.
- Dates: shown as DD/MM/YYYY or DD MMM YYYY.
- Time zone: the site's local time. Audit entries record the time to the second.
- Units of measure: each item carries its own unit - pieces, boxes, rolls, cans, metres, kilograms. There is no automatic conversion: an item is issued in the same unit it was received in. If a material arrives in boxes and is used in pieces, record the conversion in the item's description and keep one unit for both directions.

2. Roles, Permissions & Page Access

2.1 Role hierarchy

Roles are ranked, but the ranking is not a straight line - two pairs sit side by side at the same level and differ only in what they are scoped to.

Level	Role	Scoped to
0	Store Keeper	One site
1	Warehouse User	One warehouse
1	Supervisor	One site
2	Head of Department	One site
3	Logistics	All sites
3	Auditor	All sites, read-only
4	Admin	Everything

A higher rank does not open a lower role's workspace. The Logistics Portal, Warehouse Portal, Entry Log, HOD Portal, Supervisor Portal, Material Estimator and Man-Hours pages are locked to their own role exactly - Logistics outranks a Head of Department but still cannot open the HOD Portal. Admin is the single exception and reaches every workspace deliberately, for support.

2.2 Page access matrix

Page	Store Keeper	Warehouse User	Supervisor	HOD	Logistics	Auditor	Admin
Live Dashboard	[X]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]
Entry Log	[OK]	[X]	[X]	[X]	[X]	[X]	[X]
HOD Portal	[X]	[X]	[X]	[OK]	[X]	[X]	[X] (hidden - uses Admin)
Logistics Portal	[X]	[X]	[X]	[X]	[OK]	[X]	[OK] (admin)
Warehouse Portal (NEW)	[X]	[OK]	[X]	[X]	[X]	[X]	[OK] (admin shadow, picks WH)
Admin	[X]	[X]	[X]	[X]	[X]	[X]	[OK]
Reports	[X]	[OK]	[OK]	[OK]	[OK]	[OK] (read + download only)	[OK]

Records (Inventory, ledgers, POs, PRs)	[X]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]
Supervisor Portal	[X]	[X]	[OK]	[X]	[X]	[X]	[OK] (admin
Material Estimator	[X]	[X]	[X]	[OK]	[X]	[X]	flow) [OK]
Man-Hours (NEW)	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]
Auditor view	[X]	[X]	[X]	[X]	[X]	[OK]	[OK]

> Exact-role locks: Entry Log, HOD Portal, Supervisor Portal, Logistics Portal, Warehouse Portal, Material Estimator and Man-Hours are locked to their own role - a higher-ranked role does not inherit access. Admin reaches them for support purposes.

2.3 Site scope by role

Role	What they see
Store Keeper	Their own site only - they cannot view another site's stock.
Warehouse User (NEW)	Their own warehouse only - POs assigned to them, DNs they've prepared, items received from vendors. Tied to users.Warehouse_ID.
Supervisor	Their own site only - Reports, Live Dashboard, Burn Rate all site-locked.
HOD	Their own site only - but they can REQUEST material from
Logistics (NEW)	All sites globally for PRs and POs they manage. No site lock - they sit above the site boundary.
Auditor (NEW)	All sites globally, read-only. Sits at level 3 so it is not site-locked - an auditor pinned to one site could not audit. It
Admin	All sites + all warehouses globally - has the "All Sites" filter

when

shadowing the Warehouse Portal.

2.4 Default seeded accounts

The first time the app starts, these accounts are created. Change the passwords immediately.

Username	Password	Role
admin	admin2026	Admin
hod	hod2026	HOD

Username	Password	Role
supervisor	super2026	Supervisor
worker	floor2026	Store Keeper

No default logistics or warehouse_user accounts are seeded. Both roles are strictly admin-created - go to Admin Portal -> Users -> Add user. This is intentional: the procurement chain has commercial visibility (Logistics sees prices, Warehouse routes inventory) and seeded credentials would be a security liability. When you create a warehouse_user, set their Warehouse_ID to one of the values from Admin Portal -> Master DB Editor -> warehouses table - without it, the user lands on the Warehouse Portal and sees an error telling them to ask Admin.

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Username text box - your assigned username (case-sensitive).
- Password text box - masked input. Min 8 characters policy enforced on creation.
- Sign In button - checks your credentials and takes you straight to the first page your role can open.
- "Don't have an account? Request access" link - opens a self-service registration form.

Registration form elements:

- Username - must be unique
- Password / Confirm Password
- Role requested - selectbox: Store Keeper / Supervisor / HOD (Admin role cannot be self-requested)
- Site_ID - which site to associate the account with
- Phone Number - used for WhatsApp alerts (format: +966 5X XXX XXXX)
- Submit Request button - creates a pending_users row + audit log + WhatsApp alert to all admins

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v3.0.0
Role card	Your username + role badge	Color-coded: grey=Store Keeper, emerald=Warehouse, blue=Supervisor, indigo=HOD, sky=Logistics, gold=Admin
Notifications bell (NEW v3.0)	Unread count + inbox dialog	Red badge if N>0, primary button changes to "Open inbox (N unread)". Modal shows recent procurement events with mark-read controls. See §3.6
"Navigate to:" radio	Page picker	Only shows pages allowed by your role
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below
Bug/Feature reporting	Opens a dialog	See §3.4
Theme toggle	Dark / Light mode switch	Persists per-session

Element	Purpose	Notes
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
render_brand_header	Live Dashboard, Entry Log	Gold subtitle accent + current date
render_brand_header_hod	HOD Portal	Purple subtitle accent + current date
render_brand_header_admin	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

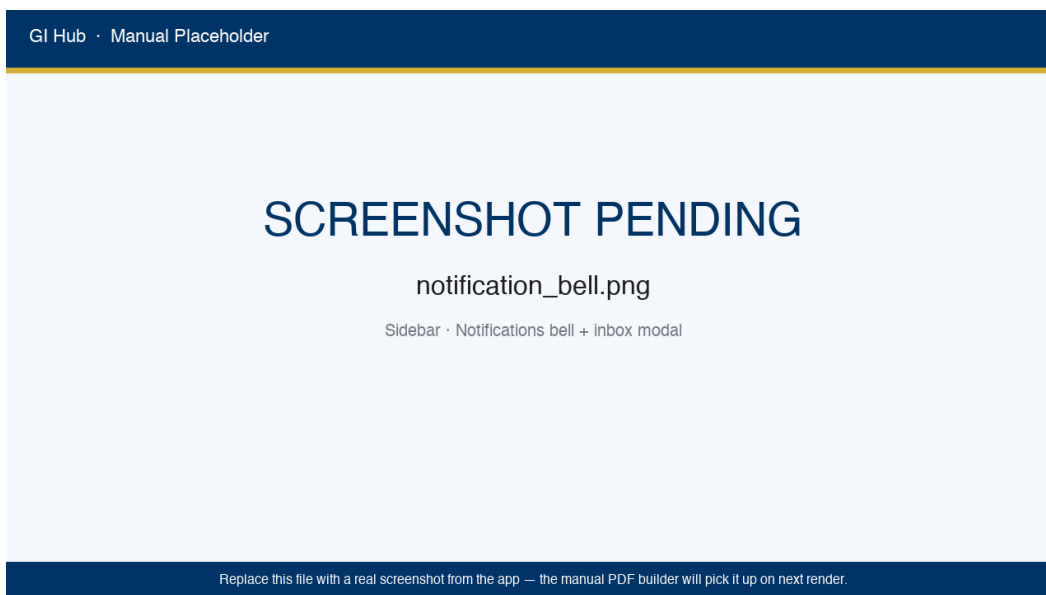
3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

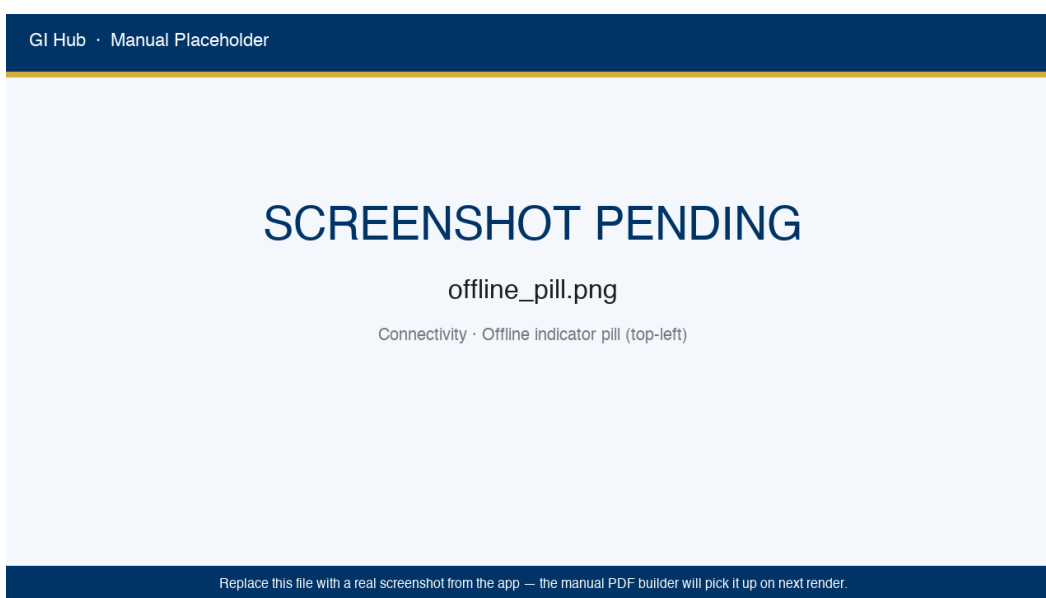
- Type selectbox - Bug Report / Feature Request
- Page dropdown - which page the issue/idea relates to (Live Dashboard, Entry Log, HOD Portal, Admin Portal, Reports, Other)
- Description textarea - up to 200 characters
- Submit button - writes to bug_reports table with user, timestamp, type, page, description. Visible to Admin in the Admin Portal Reports & Bugs tab for triage.

3.6 Notifications bell (NEW in v3.0)

Sits between the role card and the navigation radio. Every signed-in user sees their own personalised inbox of procurement-chain events.



Sidebar - Notifications bell with unread badge + inbox modal



Connectivity - offline indicator pill appears top-left when the browser loses network (Phase 7E)

The button:

- No unread: "Open inbox" (secondary button, no badge)
- 1+ unread: "Open inbox (N unread)" (primary button, red pill badge with the count, capped at "99+")

The inbox dialog (click the button):

- Only unread toggle - on by default. Flip off to see your full history.
- Per-notification card - colour-coded left border by severity:
 - Critical (red) - force-closures, T-0 delivery reminders
 - Warning (amber) - T-1/T-2 reminders, reschedule requests
 - Success (green) - DN approved / received successfully
 - Info (blue) - PR submitted, PO issued, assignments
- Per-row Mark read - flips just that notification
- Bulk [OK] Mark all as read - flips every visible row (respecting the role + site + warehouse scope)

you'd see normally)

What gets sent here:

Event you'll see	Triggered by
New PR from a site (Logistics only)	Site HOD pressing " Submit PR(s) to Logistics"
PO issued (Site HOD)	Logistics creating a PO against your site's PR
PO assigned to warehouse (Warehouse only)	Logistics routing a PO to your WH
DN awaiting your approval (HOD)	Logistics approving a DN delivery date
Incoming DN ready to receive (SK)	HOD approving DN content
Reschedule requested (Logistics)	Warehouse or HOD asking to push a date
Reschedule decided (requester)	Logistics approve/reject
Force-closure (Admin + originating HOD)	Logistics force-closing a PR/PO/line
Vendor return raised (Logistics)	Any role raising a return
Delivery reminder T-2 / T-1 / T-0	The daily sweep job (see §9)

The bell is tolerant - if a notification helper errors, the badge silently shows 0 instead of crashing the sidebar. In-app notifications ALWAYS fire; the WhatsApp side is gated by toggles in config.WHATSAPP_TRIGGERS (see §9).

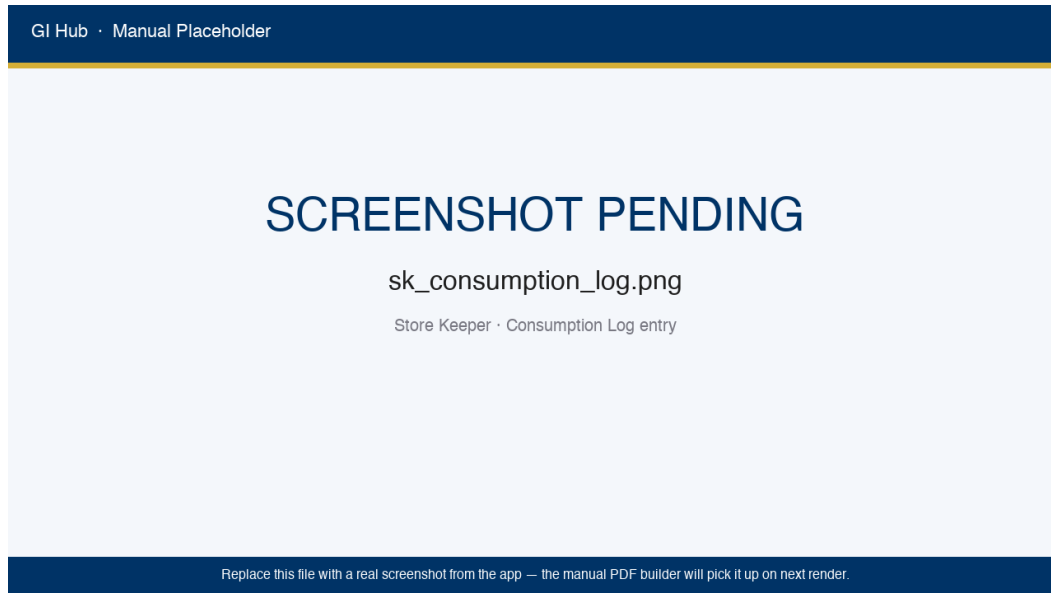
3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

> [!] OVERDUE ITEMS - Action Required: N borrowed item(s) past expected return: <item names>. Go to the Returnable Items tab to follow up.

4. Store Keeper Manual

The Store Keeper is the warehouse-floor operator. They see only the Entry Log page and the sidebar shell.



Store Keeper Consumption Log entry - main daily workflow

4.1 Pages visible

- Entry Log (only)

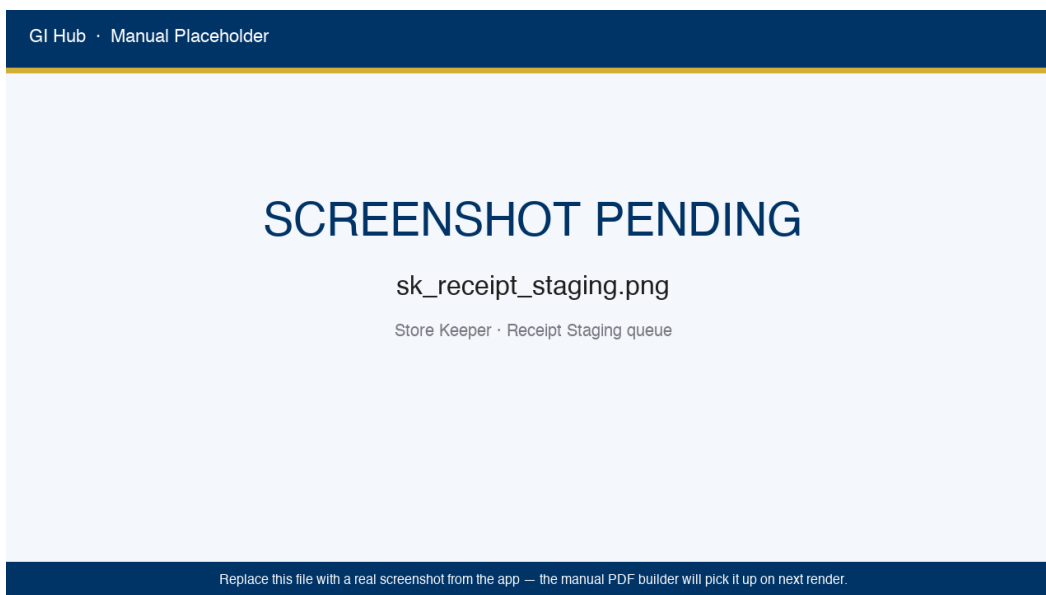
4.2 Entry Log - Tab structure

The Entry Log has four tabs:

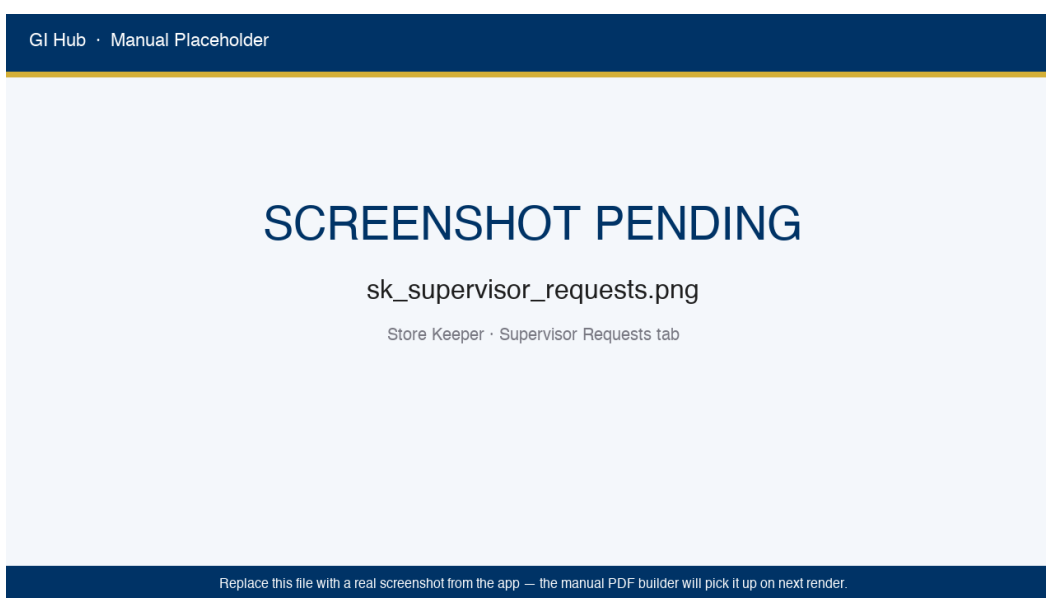
1. Consumption Log - record material issued out
2. Receipt Staging - record material arriving in
3. Returnable Items - track tools/items temporarily issued
4. Stock Count - submit physical-count reconciliations

4.3 Entry Log -> Consumption Log

This is where every material consumed by site operations is recorded.



Receipt Staging queue - review incoming materials before submitting to HOD



Supervisor Requests tab - approve / reject worker material requests

4.3.1 Top section: Bulk OCR upload (expander)

Upload Handwritten Consumption List (OCR) - for when you have a handwritten list to bulk-stage instead of typing each row.

Element	Purpose
Input method radio	Switch between "Image upload (vision AI)" and "Paste text"
File uploader	Upload PNG/JPG/JPEG/WEBP of handwritten list (vision model reads it)
Extract rows from image button	Triggers vision AI extraction -> preview rows
Paste text area	Alternative - paste tab/comma/pipe-separated rows
Parse pasted rows button	Parse the textarea
OCR Review grid	Editable preview of extracted rows with ambiguous-match pickers

Element	Purpose
Confirm All & Stage button	Pushes all reviewed rows into pending_issues

4.3.2 Mobile camera barcode/QR scanner (expander)

Barcode / QR Scanner - opens the browser camera and reads SAP codes off labels.

Element	Purpose
Camera feed	Live preview (requires camera permission)
Manual SAP code entry text box	Fallback if camera unavailable - also accepts pasted scanner output
Detected code display	Shows the last successfully scanned code in green

When a code is scanned, the material selectbox auto-populates so you don't have to search.

4.3.3 Scan / Add New Item to Queue (main expander, expanded by default)

This is the workhorse panel. Reading top to bottom:

A. Recently-scanned ring buffer

Element	Purpose
Recent: pills	Up to 5 quick-tap buttons showing your last 5 items (SAP code + truncated description). Tapping one auto-fills the material selectbox.

B. 1. Select Material

Element	Purpose
Search by SAP Code or Description selectbox	Type-ahead search across the inventory master. Format: [SAP_Code] Description.

When you pick a material, two cards appear automatically:

C. Item Snapshot card

A compact dark card showing:

- SAP code (gold monospace)
- Material description (bold)
- 3-stat strip:
 - SITE STOCK (color-coded: red=empty, amber=below min, green=ok)
 - 30-DAY BURN (units consumed last 30 days)
 - DAILY RATE
- Inline 70x22 SVG sparkline - last 30 days consumption trend
- Status badge bottom-right (OK / Low / Below Min / Empty)

D. FEFO panel (lot suggestion)

If lots exist for this item at your site:

Element	Purpose
FEFO - First Expiry, First Out header (amber)	Banner
Per-lot row	Lot number (monospace) - xqty (dim) - "Exp <date>" (amber if <90d, red if expired) - USE FIRST badge on the first row

The system will automatically attach the top-FEFO lot number to your consumption row when you submit.

E. Pull from a different lot (FEFO override) - expander

Only renders when 2 or more open lots exist for this item at your site.

Element	Purpose
Lot to pull from instead selectbox	Default: "Keep FEFO suggestion". Other options: all other open lots with expiry + remaining qty shown
Reason for override text box (200 char max)	Mandatory. Min 5 characters to activate the override
Amber confirmation banner	Appears once override is active: "FEFO override active: pulling from <chosen> instead of <suggested>"

When activated:

- Your consumption row gets Lot_Number = chosen + FEFO_Override = reason
- Audit log entry FEFO_OVERRIDE is written with full context
- A WhatsApp alert is queued to the site HOD in real time

F. 2. Fill Entry Details (dynamic form)

Fields appear based on the pending_issues schema. Every required field is marked with *.

Field	Type	Purpose
Date	Date picker (default: today)	When the consumption happened
Quantity	Number input (min 0.1)	How much you're issuing. Above this field is a stock badge showing current site stock; a red [!] warning appears if you type more than available.
Work_Type	Selectbox (sourced from system_settings.Work_Type)	Classification: PM Work, Breakdown, Project, Shutdown, Inspection, etc.
Issued_By	Text (auto-filled with your username)	Who handed it over
Issued_To	Text (smart-defaults from your last entry)	Recipient personnel/department
Tank_No	Text	Equipment tag if applicable
Serial_No	Text (optional)	Item serial number if tracked
PR_Number	Text (smart-default)	Linked purchase request if applicable
Remarks	Text (optional)	Free-form notes

Hidden side-effects:

- Site_ID is auto-set to your site
- Lot_Number is auto-set to FEFO top suggestion (or your override)

status is set to 'draft' until you submit

G. **[!] Override Expiry Warning (amber card with checkbox)**

If the system detects a short-dated batch at your site for this item, it will hard-block the Add to Grid button until you check this box, confirming you've physically pulled from the expiring batch first.

H. **Add to Grid button**

Validates that:

- A material is selected
- No required fields are empty
- Quantity <= current site stock (over-issue guard)
- Expiry override is checked when needed

On success: inserts a pending_issues row with status='draft', toasts [OK] Added to staging queue, reruns the page.

4.3.4 Staging Queue section

Element	Purpose
Staging Queue header + count badge	Shows how many draft rows you've accumulated
Data editor table	Edit any row inline before submitting
Save Draft Edits button	Persists edits in-place (rows stay as drafts)
Submit Grid to HOD button	Flips all draft rows to status='pending_hod' + queues a WhatsApp alert to the site HOD with the full item list

After submission, the rows are locked from your view - they appear in HOD Portal -> EOD Commit tab for approval.

4.4 Entry Log -> Receipt Staging

For when materials arrive at your site.

4.4.1 Upload Delivery Note (OCR) - expander

Identical lane structure to consumption OCR - upload an image of the delivery note OR paste text. Vision AI extracts rows for bulk staging.

4.4.2 Add Receipt to Queue - expander (expanded by default)

A. 1. **Select Material**

Element	Purpose
Search by SAP Code or Description selectbox	Same type-ahead search as consumption

When picked, a blue-tinted info card shows:

- Mat Code

—

UOM

B. 2. Fill Receipt Details (dynamic form)

Field	Type	Purpose
Date	Date picker (today)	Delivery date
Quantity	Number input	Units received
Supplier	Text (optional)	Vendor name
Lot_Number	Text (optional)	If blank + Expiry_Date is set, system auto-generates LOT-YYYYMMDD-SAP
Expiry_Date	Date picker (optional)	Lot expiry - when set, triggers lot master entry
PR_Number	Text (optional)	Link to an open Purchase Request
Serial_No / Vehicle_No / DN_No / Pallet_No / Mob_From / Mob_To / Prepared_by / Driver_Name	Text (optional)	Logistics tracking fields
Remarks	Text	Free notes

C. Add to Receipt Queue button

Validates: material picked, required fields populated. Inserts a pending_receipts row with status='draft'.

4.4.0 Incoming Delivery Notes from Warehouse (NEW in v3.0)

A new expander appears at the TOP of the Receipt Staging tab. It's only populated when the Warehouse has prepared a DN bound for your site AND your HOD has approved it. If empty, the expander shows: "Nothing inbound. Logistics -> HOD-approved DNs will appear here for you to confirm physical receipt."

When DNs are inbound, each appears as its own container:

Element	Purpose
DN header line	DN <number> - PO <number> - Warehouse <id> - DN Date
Line count + total qty	Alert: how many SKUs, how many units total
View lines expander	Material_Code, Description, Qty, UOM, Lot_Number, Expiry_Date, Remarks - read-only preview of every line
[OK] Mark as Received button	Confirms physical receipt. One receipt is recorded per delivery line, each keeping its delivery note, warehouse and originating order so the trail is complete; the delivery closes

When to use:

- The Warehouse delivered the materials physically to your site
- HOD already approved the DN content (you'll see it appear without any action)
- You've inspected the goods and they match the DN qty + lot

When NOT to use:

—

Direct deliveries from supplier to your site (those go through the existing Add Receipt to Queue -> Submit to HOD flow below; the email-driven PR/PO path remains supported)

- Partial receipts (the current flow assumes you confirm the full DN qty - for partials, raise a Vendor Return on the diff after confirming and ask your HOD)

This is the FINAL step in the procurement chain. After you click Mark as Received:

- Logistics sees dn_received_by_sk in their notifications
- Warehouse sees the same
- The PO and PR move toward closure if this DN completes the order

4.4.3 Receipt Draft Queue section

Element	Purpose
Receipt Draft Queue header + count badge	Number of drafts you've accumulated
Data editor table	Inline edit before submit
Save Draft Edits button	Persists
Submit to HOD for Approval button	Flips drafts to status='pending_hod' + WhatsApp alert to HOD with item list

4.5 Entry Log -> Returnable Items

For tools, gauges, fittings temporarily handed to personnel - items that should come back.

4.5.0 Smart Scan workflow (new - 2026-06)

A two-step camera flow sits at the top of the tab. Use it to skip typing borrower + tool details by hand; the existing manual form still works exactly as before and stays visible underneath.

Step 1 - Scan the employee badge

1. Click into the Smart Scan (Beta) expander.
2. Hold the borrower's printed badge in front of the camera. (Badges are generated in Admin Portal -> Employees -> Roster + Badges - see §7.6.)
3. The system decodes the QR (which carries only the ID_Number, no PII), looks it up in the Employees master, and shows:
 - [OK] Green success card - EMP-1042 - Ahmed Ali (+9665...) - the borrower's name + phone are now staged.
 - [X] Red error card - Badge couldn't be read OR the employee is inactive/suspended. Re-snap or use the manual form.

Step 2 - Scan the tool

1. After a successful badge scan, a second camera input appears. Point it at the tool.
2. The active YOLO model (managed in Admin Portal -> Tool Catalogue, see also docs/cv_training_guide.md for how the model gets trained and promoted) classifies the image. Three outcomes:

-

Confidence ≥ 0.75 -> [OK] Auto-fill. The detected tool name + borrower details flow into the manual form below; review and click "Issue Item".

- Confidence 0.30-0.74 -> [!] A "Top candidates" radio appears. Pick the right one and click "Use this tool". The form pre-fills.
- No active CV model OR confidence < 0.30 -> An info banner explains the fallback. Borrower fields stay pre-filled; type the tool name into the manual form yourself.

Click "Start over (clear scan)" any time to reset and scan a different borrower.

What happens if the AI is wrong

You're never locked in. Every Smart Scan write-through goes into the manual form's fields - Name, Phone, Material name. Edit anything that's wrong before clicking "Issue Item". The submitted record stores `cv_detected=1` and `cv_confidence` so admin reports can later show adoption + accuracy telemetry. Manually-corrected loans don't get tagged as CV - that's intentional, see §10 for the data model details.

Automatic WhatsApp reminders (Phase 6E)

Once a loan is in the system, the WhatsApp worker fires four reminders automatically. The cadence and recipient list is:

Offset	Severity	In-app	WhatsApp
T-2h (2 hours before due)	info	Site SK badge	Borrower
T-0 (due now)	warning	Site SK badge	Borrower
T+2h (2 hours overdue)	warning	Site SK badge	Borrower + every Site SK
T+24h (24 hours overdue)	critical	Site SK + Supervisor badges	Borrower + every Site SK + every Site Supervisor

The borrower's phone is resolved first from the Employees master (CV-scanned loans) and falls back to the manually-typed `borrower_phone` column for legacy/manual loans. The system de-dupes per loan + offset so restarting the worker mid-day never causes double-fires. Admins can mute any of the four events via `config.WHATSAPP_TRIGGERS` if a particular escalation gets too noisy.

4.5.1 Issue a Returnable Item - expander (expanded by default)

Field	Type	Purpose
Material / Tool Name	Text	What you're handing out
UOM	Text (e.g., Pcs, Set)	Unit
Quantity	Number (min 0.1)	How many
Borrower Name	Text	Person taking custody
Borrower WhatsApp Number	Text (optional, +966 ...)	For overdue alerts
Expected Return Date	Date (default: tomorrow)	When you expect it back
Expected Return Time	Selectbox: 04:15 PM, 06:15 PM, Custom Time...	Time-of-day expectation
Issue Item button	Validates name+borrower, writes to <code>returnable_items</code>	

4.5.2 Currently Borrowed Items section

Shows all borrowed (not yet returned) items at your site.

- Overdue red banner - when one or more items are past their expected return time
- Borrowed items table (styled HTML) with columns: ID, Material, UOM, Qty, Borrower, Phone, Given Time, Expected Return, Status (pill: green "On Loan" / red "[!] Overdue")
- Mark as Returned section:
- Selectbox of borrowed items
- [OK] Mark as Returned button - updates status='returned', busts cache, toast confirms

4.6 Entry Log -> Stock Count

For reconciling physical shelf count with system stock when they don't match (damage, expiry disposal, miscount, found-extra).

4.6.1 1. Select Material to Count

Element	Purpose
Search by SAP Code or Description selectbox	Pick the material to reconcile
Help caption	Explains when to use this tab

4.6.2 2. Enter Count Details (only renders after picking a material)

Left column

Element	Purpose
System Qty card	Read-only dashboard showing current stock at your site (auto-fetched)
Counted Qty input	What you actually count on the shelf right now (defaults to system qty)
Variance preview banner	Color-coded: green " Found N extra" or red " Short by N" or grey "No variance"

Right column

Element	Purpose
Reason Code selectbox	9 options. Default depends on variance direction.
Notes textarea (300 char max)	Free explanation: "found behind shelf 3", "damaged in transit", etc.

Reason codes:

Code	Label
cycle_count	Cycle count correction
damaged	Damaged / unusable
expired_disposal	Expired - disposed
miscount_in	Miscount - found extra

Code	Label
miscount_out	Miscount - short
lost	Lost / unaccounted
theft	Suspected theft
return_to_supplier	<- Returned to supplier
other	Other (see notes)

4.6.3 Submit section

Element	Purpose
Amber warning banner	"Submitting this sends the count to your HOD for approval.
Submit Count for HOD Approval button	Disabled while generating the approval process: writes to stock_adjustments with status='pending_hod', queues WhatsApp to site HOD with count details, audit-logs.

4.6.4 Recent Adjustments at This Site (history)

Read-only table showing: ID, SAP Code, Material Name, Variance, Reason, Status, Submitted By, Submitted At.

4.7 Store Keeper - Use Cases

Use Case 1: Issue 10 Pipe Gaskets to PM Work

1. Log in -> Entry Log -> Consumption Log tab
2. Open Scan / Add New Item to Queue
3. Search "Pipe Gasket" -> pick from dropdown
4. Read the stock badge (e.g., " 156 Pcs"), confirm FEFO lot suggestion
5. Quantity: 10
6. Work_Type: PM Work
7. Issued_To: name of recipient
8. Tank_No / PR_Number: as applicable
9. Click Add to Grid
10. Repeat for any other items in this batch
11. Review the Staging Queue table -> edit if needed -> click Submit Grid to HOD

The row(s) are now in HOD's EOD queue. Stock won't decrease until HOD commits EOD.

Use Case 2: Receive a delivery of 200 units with expiry date

1. Entry Log -> Receipt Staging tab
2. Open Add Receipt to Queue
3. Pick material, fill Quantity (200), Supplier, Expiry_Date (e.g., 2027-06-11)
4. Lot_Number: leave blank (system will generate LOT-20260611-<SAP>) OR type the supplier's lot ID

5.

PR_Number: link if applicable

6. Click Add to Receipt Queue
7. Click Submit to HOD for Approval when done staging

Use Case 3: Found 5 extra units of an item (physical count > system)

1. Entry Log -> Stock Count tab
2. Pick the item from the dropdown
3. Read the System Qty card (e.g., "95")
4. Counted Qty: 100
5. The variance banner turns green: " Found 5 extra"
6. Reason: Miscount - found extra
7. Notes: e.g., "Found in box behind shelf 3"
8. Click Submit Count for HOD Approval

HOD will approve in HOD Portal -> Adjustments. After approval, your live stock matches reality and a synthetic receipt row of +5 is posted to the ledger.

Use Case 4: Issue a tool to a borrower

1. Entry Log -> Returnable Items tab
2. Issue a Returnable Item
3. Material: Torque Wrench 1/2", UOM: Pcs, Qty: 1
4. Borrower Name, WhatsApp Number, Expected Return: today 06:15 PM
5. Click Issue Item

If the item is not returned by 06:15 PM, the borrower (and you) get a WhatsApp alert.

Use Case 5: FEFO override (the right bin is blocked)

1. Entry Log -> Consumption Log -> pick material with 2+ open lots
2. Scroll to FEFO panel - note the suggested lot (earliest expiry)
3. Open Pull from a different lot expander
4. Pick a different lot from the dropdown
5. Type a reason >= 5 chars (e.g., "FEFO bin blocked by pallet, clearing 1700")
6. Amber confirmation banner appears
7. Fill quantity + other fields
8. Click Add to Grid

Your HOD gets a real-time WhatsApp alert documenting the override.

4.8 Store Keeper - FAQ

Q: I can't see Live Dashboard, HOD Portal, or Reports. Did I lose access? A: No - Store Keepers only have access to Entry Log by design. If you need to view stock, ask your Supervisor or HOD.

Q: I submitted to HOD by mistake. Can I cancel? A: Not directly. Contact your HOD - they can reject the

row in their EOD Commit review.

Q: The "Add to Grid" button is disabled or the form rejects my entry. A: Check three things: (a) is quantity > current stock? (b) is the expiry override needed but unchecked? (c) are all required * fields filled?

Q: I tried to issue 20 but the system says only 15 available. A: Believe the system. Either physically recount, or if there is genuinely 20 and the system disagrees, use the Stock Count tab to log a +5 adjustment.

Q: I don't have a barcode scanner. How do I quick-find items? A: Type a partial SAP code or description in the selectbox - it has type-ahead search. Or use the Recently-scanned pills if you've used the item recently.

Q: WhatsApp alerts aren't reaching the HOD when I submit. A: HOD's phone number must be set in their user profile. Tell the HOD to update it via Admin (or check with Admin directly).

Q: My borrowed item is overdue but I haven't received an alert. A: WhatsApp alerts are sent by a background service. Ask your Admin to confirm it is running - queued messages send as soon as it is back, so nothing is lost in the meantime.

Q: I uploaded an OCR image and it extracted wrong items. A: The OCR review grid lets you fix every row before staging. Pick the correct material from the suggested-matches dropdown, or type it manually.

Q: How do I see my recent activity? A: Submitted/approved items don't show on your screen. Ask HOD or Admin to filter the audit log by your username.

10. Data Model & Concept Reference

10.1 Core movement tables

Table	What it stores	Identity-math role
inventory	Master catalogue: SAP_Code (PK), Description, Material_Code, UOM, Minimum_Qty, Unit_Cost	The "items that exist" - defines what can be moved
receipts	Every received unit (post-commit). Includes Lot_Number, Expiry_Date, Supplier, PR_Number, Unit_Cost	+ adds to stock
consumption	Every consumed unit (post-EOD-commit). Includes Lot_Number, FEFO_Override, Work_Type	- subtracts from stock
returns	Tools and equipment returned to inventory	+ adds back
pending_issues	Pre-commit staging for consumption (status: draft -> pending_hod -> approved/rejected -> committed)	- does NOT affect stock
pending_receipts	Pre-commit staging for receipts	- does NOT affect stock

Identity formula: $\text{Current_Stock} = \text{Total_Received} - \text{Total_Consumed} - \text{Total_Returned} - \text{computed live in v_site_stock view.}$

10.2 Document-type tables

Table	What it stores	Lifecycle
stock_adjustments	Physical-count reconciliations	pending_hod -> approved (posts synthetic ledger row) / rejected
requests	Cross-site material transfers	pending -> approved / rejected -> fulfilled
pr_master	Purchase Request lines (manual or PDF-extracted)	status: open / closed; workflow_state: draft -> submitted -> approved -> in_progress -> received
returnable_items	Tools temporarily issued out	borrowed -> returned / overdue
lots	Lot master metadata (FEFO source-of-truth)	open -> exhausted / expired / disposed / quarantine

10.3 Supporting tables

Table	Purpose
users	Auth + role + site + phone
pending_users	Self-registration queue
system_audit_log	Immutable activity record
system_settings	Dropdown values (Work_Type, Site)
app_settings	Key/value config (thresholds, maintenance_mode, last_backup_at)
whatsapp_queue	Outbound message queue
pwa_tokens	API tokens for offline PWA
bug_reports	User-submitted issues/ideas
report_schedules	Scheduled report definitions
report_archive	Generated report metadata

10.4a Procurement chain tables (NEW in v3.0)

Table	What it stores	State machine
warehouses	Master of receiving locations	active / inactive
vendors	Supplier master	active / inactive
purchase_orders	PO header (PO_Number UNIQUE)	open -> partially_delivered -> delivered
po_items	PO line items with rl_bl_family tag	open -> partially_delivered -> delivered
po_shipment_schedule	Parsed PO Annexure rows	pending / shipped / delivered / delayed
po_assignments	Logistics -> Warehouse routing	assigned -> acknowledged -> partial / received / closed / cancelled
delivery_notes	DN header (DN_Number UNIQUE)	DN state machine: draft -> pending_logistics -> pending_hod -> pending_sk -> received (or rejected from any pending)
dn_items	DN line items	pending / received / partial / returned /
po_returns	Vendor returns (raised by any role)	open -> vendor_acknowledged / resupplied / cancelled
po_reschedule_requests	Date-change asks	pending -> approved / rejected
po_force_closures	Force-closure audit log	(terminal - write-once)
app_notifications	In-app bell inbox	read_at IS NULL (unread) -> timestamp (read)
delivery_reminders_sent	T-2/T-1/T-0 idempotency log	(terminal - UNIQUE constraint)

10.4 Views

View	Purpose
v_live_stock	Per-SAP global stock (sum across all sites)
v_site_stock	Per-(SAP, Site) live stock - the canonical "what's at this site"
v_expiring_stock	Lots/receipts in expiry buckets
v_supplier_activity	Per-supplier receipt rollups
v_lot_balance	Per-lot Received_Qty / Consumed_Qty / Remaining_Qty (identity math)

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	Damaged / unusable	Physical damage
expired_disposal	Expired - disposed	Expired and physically discarded
miscount_in	Miscount - found extra	Found more than system shows
miscount_out	Miscount - short	Found less than system shows
lost	Lost / unaccounted	Genuinely missing
theft	Suspected theft	Deliberate removal
return_to_supplier	<- Returned to supplier	Quality issue, sent back
other	Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent
- submitted - sent to procurement
- approved - procurement greenlit
- in_progress - supplier engaged
- received - fully received

11.4 Lot statuses

- open - has remaining quantity, available for FEFO
- exhausted - fully consumed (auto)
- expired - past expiry date (manual or background)
- disposed - physically removed (manual)
- quarantine - held pending inspection (manual)

11.4a DN states (NEW v3.0)

draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received With rejected as terminal from any pending state.

11.4b PO + PO line states (NEW v3.0)

PO header: open -> partially_delivered -> delivered -> closed / force_closed / cancelled PO item line: open -> partially_delivered -> delivered -> returned / closed / force_closed

11.4c Force-closure target types (NEW v3.0)

Code	Label
pr	Whole PR closed
po	Whole PO closed
po_item	Single line on a PO closed

11.4d Reschedule + vendor return states (NEW v3.0)

Reschedule: pending -> approved / rejected Vendor return: open -> vendor_acknowledged / resupplied / cancelled

11.4e RL/BL family tags (NEW v3.0)

Tag	Meaning	Detection rule
RL	Rubber Lining	Substring RL-, RUBBER LINING, RUBBER-LINING in Material_Code OR Description
BL	Brick Lining	Substring BL-, BRICK LINING, BRICK-LINING, BRICK MATERIAL
NULL	Neither family	Default

Logic in the system. NEVER a combo string - RL takes precedence if both tokens are present.

11.4f Notification severity (NEW v3.0)

Severity	Visual	When used
info (blue)	Info pings - PR submitted, PO issued, assignments	
warning (amber)	T-2/T-1 reminders, reschedule requests, rejections	
success (green)	DN approved successfully, delivery completed	
critical (red)	T-0 reminder, force-closures, urgent escalation	

11.4g Logistics-status on PR rows (NEW v3.0)

Code	Meaning
site_draft	HOD has the PR but hasn't submitted to Logistics yet
submitted	Sitting in Logistics queue waiting for PO issuance
in_po	A PO has been issued against this PR line
closed	PR fulfilled normally
force_closed	Logistics force-closed the PR with a reason

11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	A database setting that lets many people read while one person writes. Nothing to configure - it is on
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)
Identity math	Stock derived from movements, never stored as a counter

12. FAQ - Master Index by Role

12.1 General (any role)

Q: I forgot my password. A: Contact your Admin. Admin -> Access Control -> Force Password Reset.

Q: I get "Permission denied" when clicking a page. A: Your role doesn't have access. See §2.2 for the access matrix.

Q: I want to suggest a feature or report a bug. A: Any page -> sidebar -> Bug/Feature reporting dialog. Admin reviews in Admin Portal -> Reports & Bugs.

Q: The page is blank or broken. A: Hard-refresh (Cmd+Shift+R / Ctrl+F5). If it persists, check with Admin - they can see audit logs to diagnose.

12.2 Store Keeper - see §4.8

12.3 Supervisor - see §5.5

12.4 HOD - see §6.14

12.5 Admin - see §7.13
